

An Introduction to 3D Printing

Part 3: Science

*Vibrations and Waves, Linear Motion, and Thermal
Expansion*

Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

Thermal Expansion

Vibrations and Waves

- Toolhead is attached to tensioned belts.
- Toolhead creates vibrations when it accelerates.
 - Speeds up.
 - Slows down.
 - Changes direction.
- Belts (strings) + vibrations (waves) = Physics.
 - Motions systems with belts have a **natural frequency**.
 - What does this mean?
 - *Hint:* Think of a guitar or piano string.

Repeating Mistakes

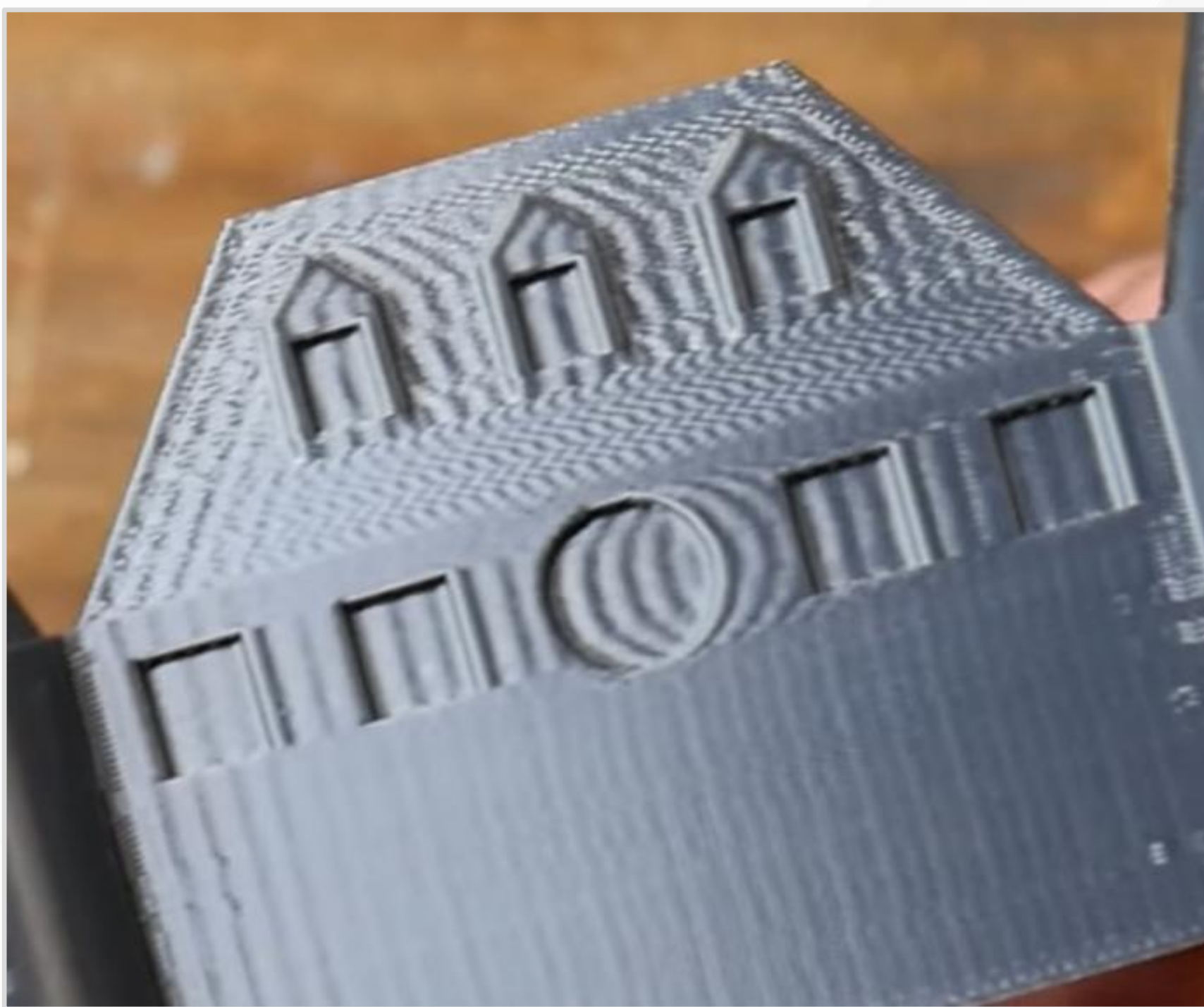
- What happens when the toolhead hits a natural frequency?
 - Resonance.
- Toolhead excites the belts (adds energy) in the form of vibrations.
- Vibrations travel up and down the belts.
- Resonance maximizes vibrations.
 - Standing waves experience constructive interference.
- By comparison, ordinary waves interfere randomly.
 - Smaller vibrations that easily dissipate.

Repeating Mistakes

- What does this mean for 3D printers?
 - Large vibrations in belts push and pull on the toolhead → Small errors in position.
- **Errors repeat based on the frequency.**
- Imagine writing on paper with someone bumping your arm at a constant time interval.

- Where do you see evidence of repeating errors?
- Sharp corners → Large accelerations → Large forces ($F = ma$).
- Which direction was the toolhead moving?
- From left to right.

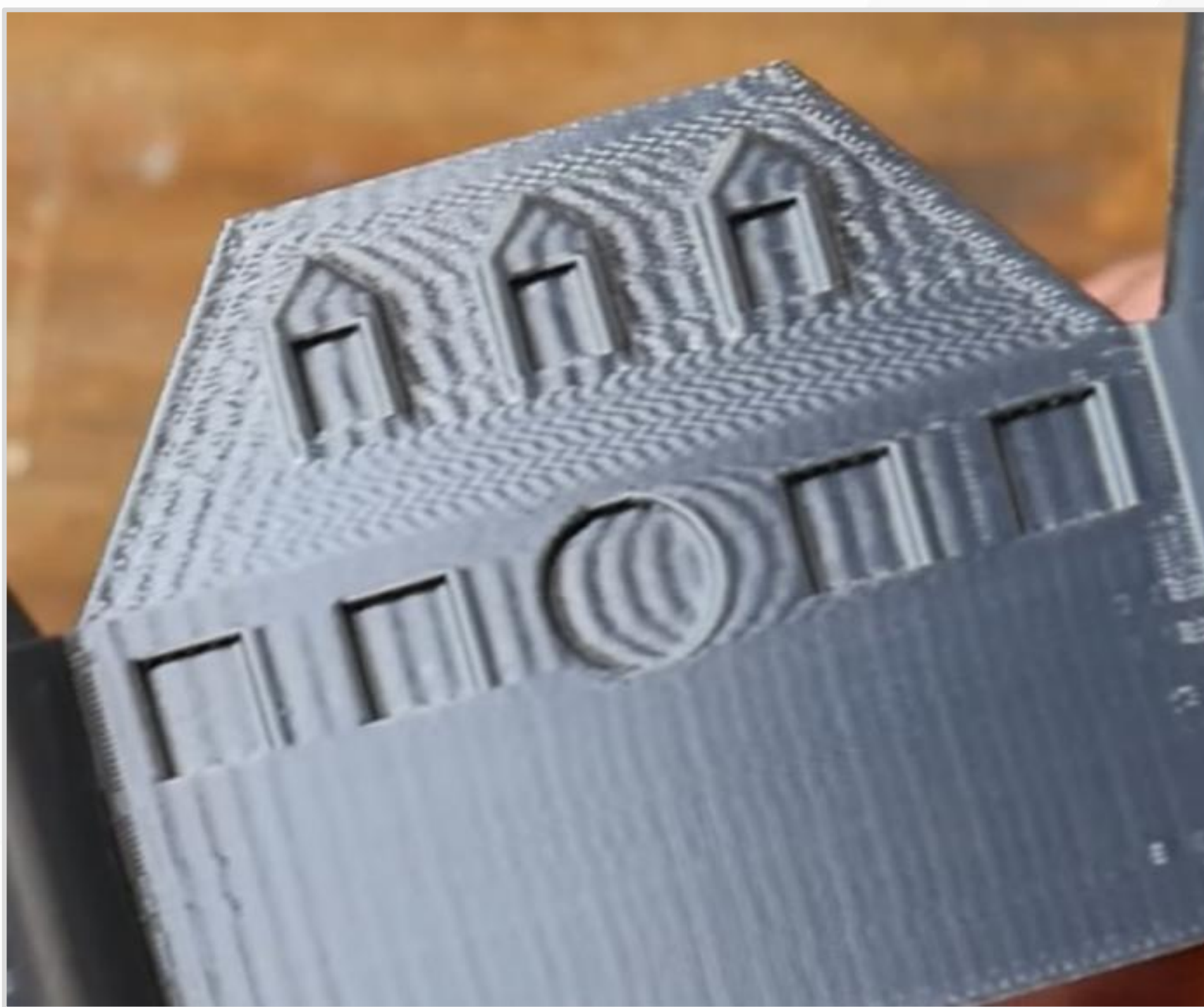
(Screenshot from
Teaching Tech)



- How can you determine frequency?
- Lookup printing speed in settings.
- Measure distance between repeating patterns.

$$freq = \frac{print_speed}{repeating_dist}$$

(Screenshot from
Teaching Tech)



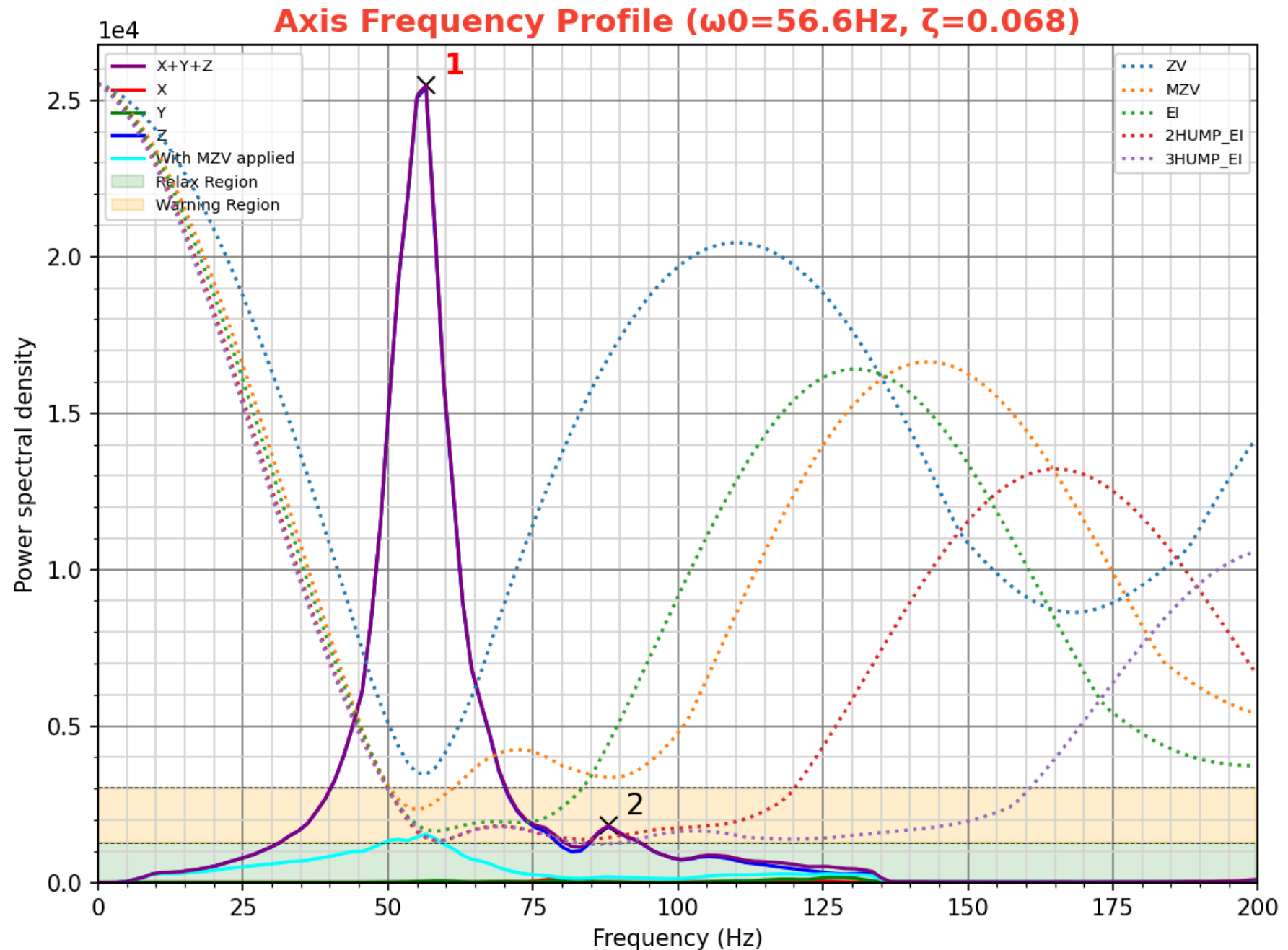
Input Shaping

- How can you fix vibrations?
- Input shaping adds extra movements that cancel vibrations.
 - Destructive wave interference.
- Similar to noise-canceling headphones.

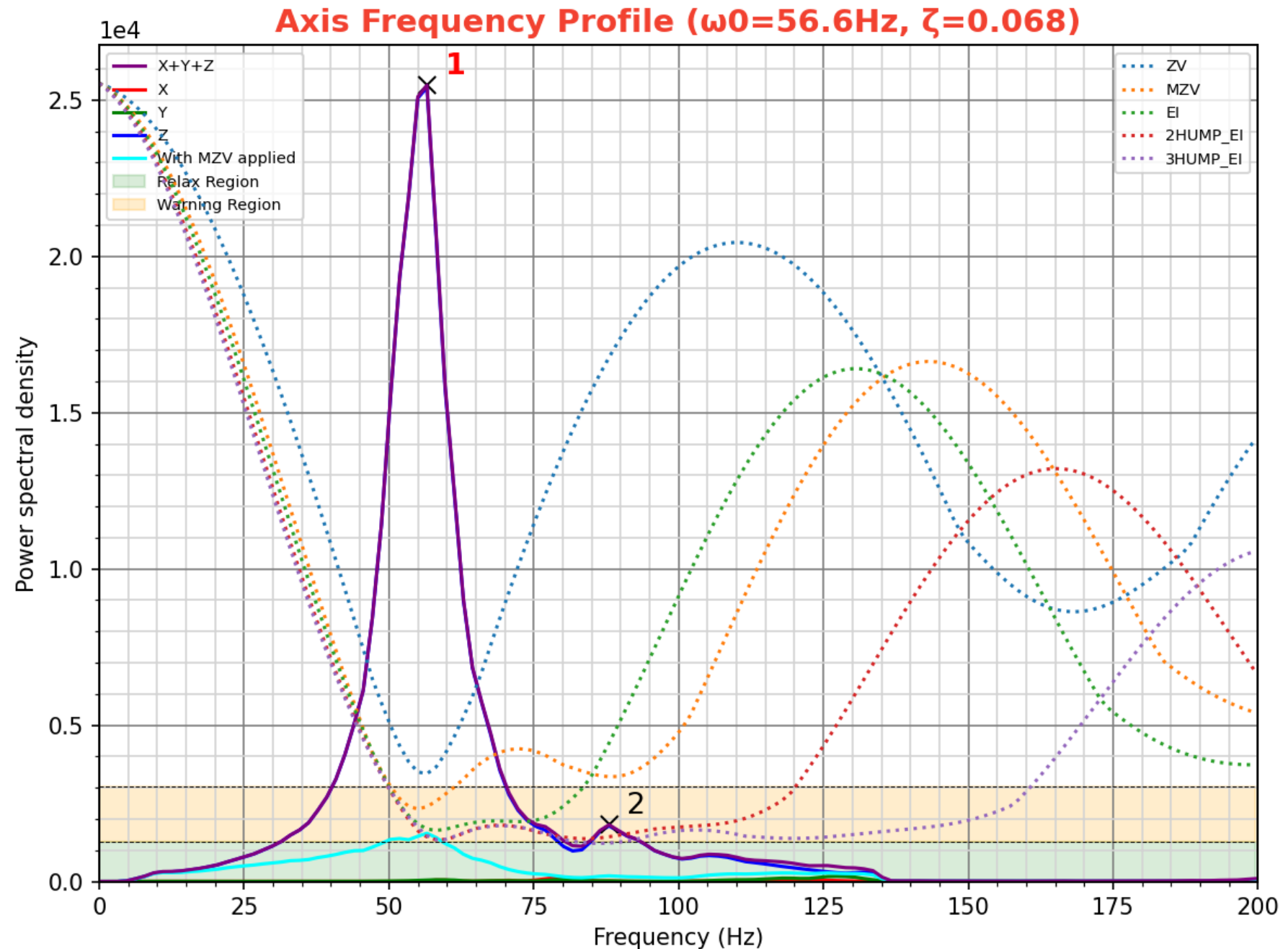
(Animation, Demo by
[ACS Motion Control](#))



- Shake and Tune Plugin.
- Attach an accelerometer to the toolhead.
- Move the toolhead back and forth at different frequencies.
- Script analyzes the results, makes this graph, and suggests Input Shaper settings.



- X-axis results from my printer.
- **Purple** → Before
Cyan → After
- Single and narrow peak is ideal.
- Each printer and axis has a characteristic shape.
- Ignore high frequencies on the right side of the graph.



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Speed Limits

I want to print faster. (Why? Gotta go fast.)

- *Acceleration*: 10,000 mm/s²
- *Max speed*: 300 mm/s
- *Typical extrusions*: 0.4 mm wide, 0.2 mm tall
- *Max volumetric flow*: 22 mm³/s
- What should I upgrade?
 - Motors? Top speed or acceleration?
 - Max flow (heater and nozzle combo)?

Velocity and Acceleration

- What is the distance to reach top speed?
 - Starting speed (cornering speed, square corner velocity, jerk). 5 mm/s.
 - Increasing cornering speed *greatly* increases vibrations.
- What is the smallest extrusion that reaches top speed?
 - 9 mm to accelerate *and* decelerate.

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$300^2 = 5^2 + 2 \times 10,000(\textit{dist})$$

$$\textit{dist} = 4.5 \textit{ mm}$$

Velocity and Flow

- Can my hotend melt enough plastic at top speed?
- *Almost, $275 < 300$.*
- I guess I need more max flow.
- What if I change nozzles?
New extrusions
0.6 mm wide, 0.3 mm tall.
- *Even worse, $122 \ll 300$.*

$$print_speed = \frac{flow_rate}{cross_section}$$

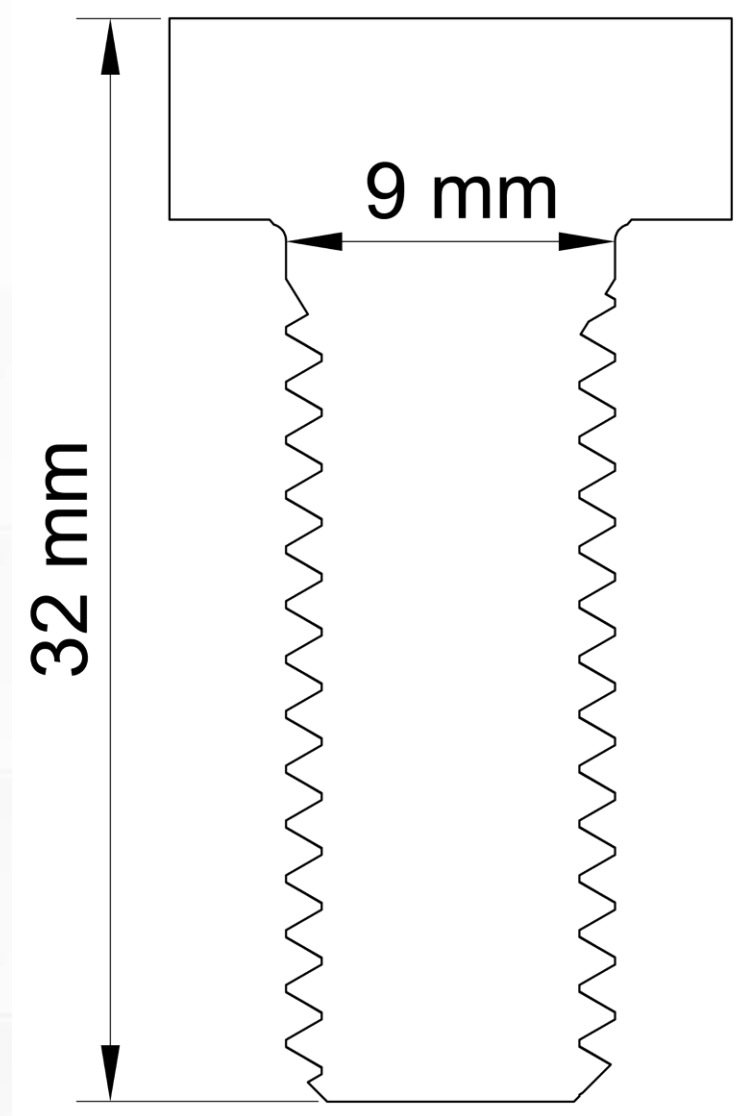
$$print_speed = \frac{flow_rate}{extrusion_width \times layer_height}$$

$$print_speed = \frac{22}{0.4 \times 0.2}$$

$$print_speed = 275 \text{ mm/s}$$

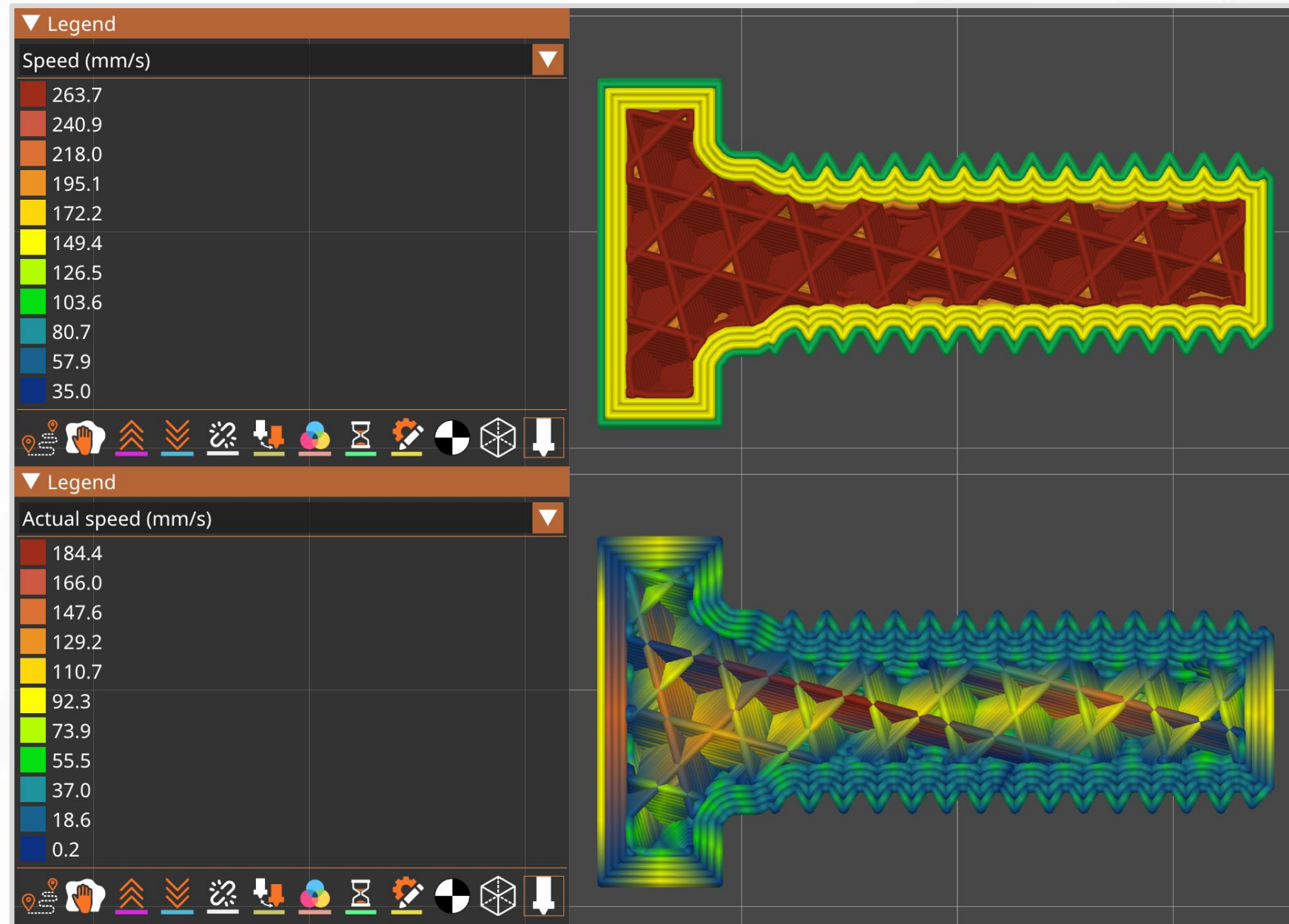
Finding Limits

- Real life limits are more complex and nuanced.
- *Example:* Printed bolt.
- What is going to limit my printer?
- Let PrusaSlicer do all the math!



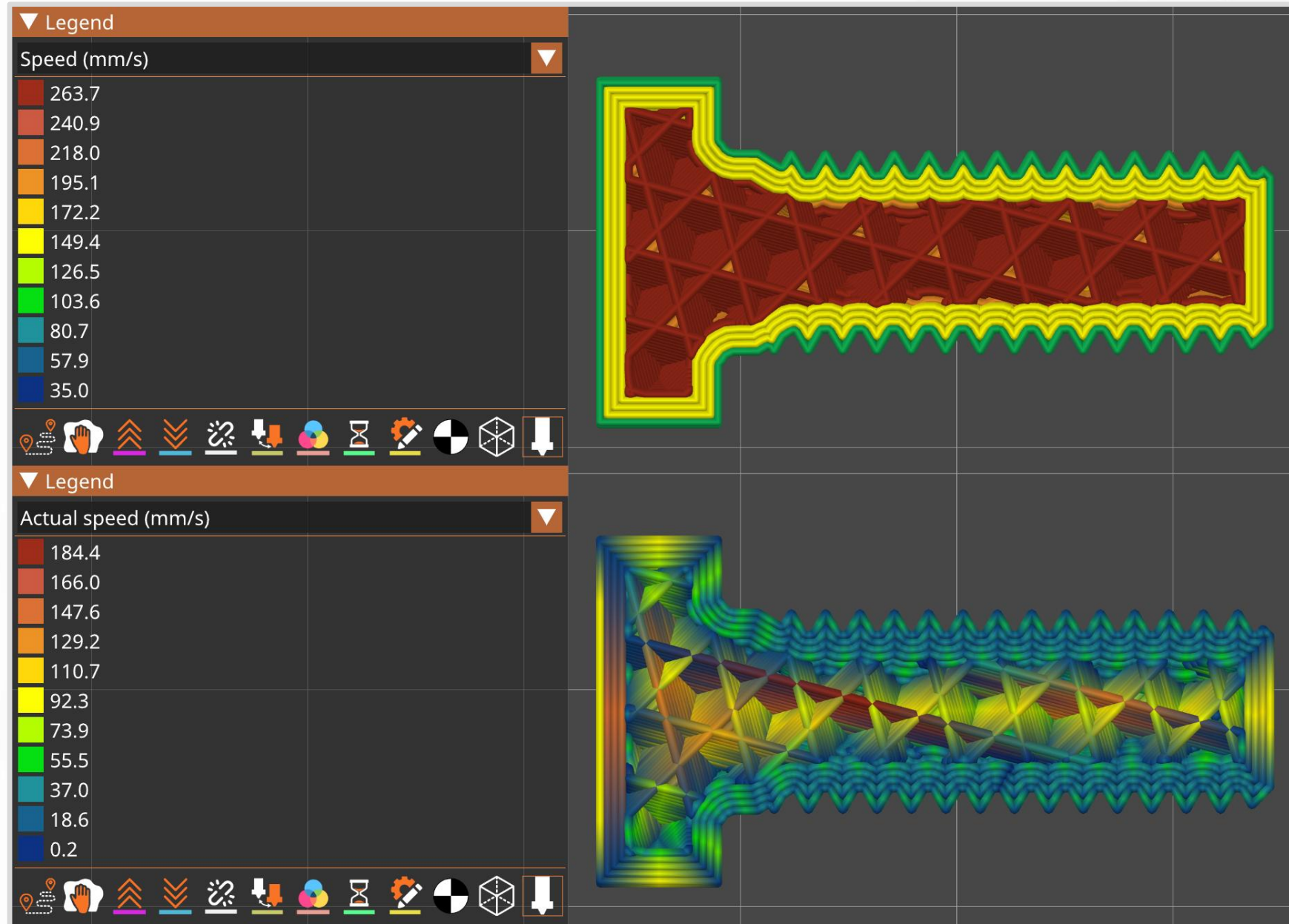
- Max requested speed:
264 mm/s
- Looks close to the estimate.

- Max actual speed:
184 mm/s



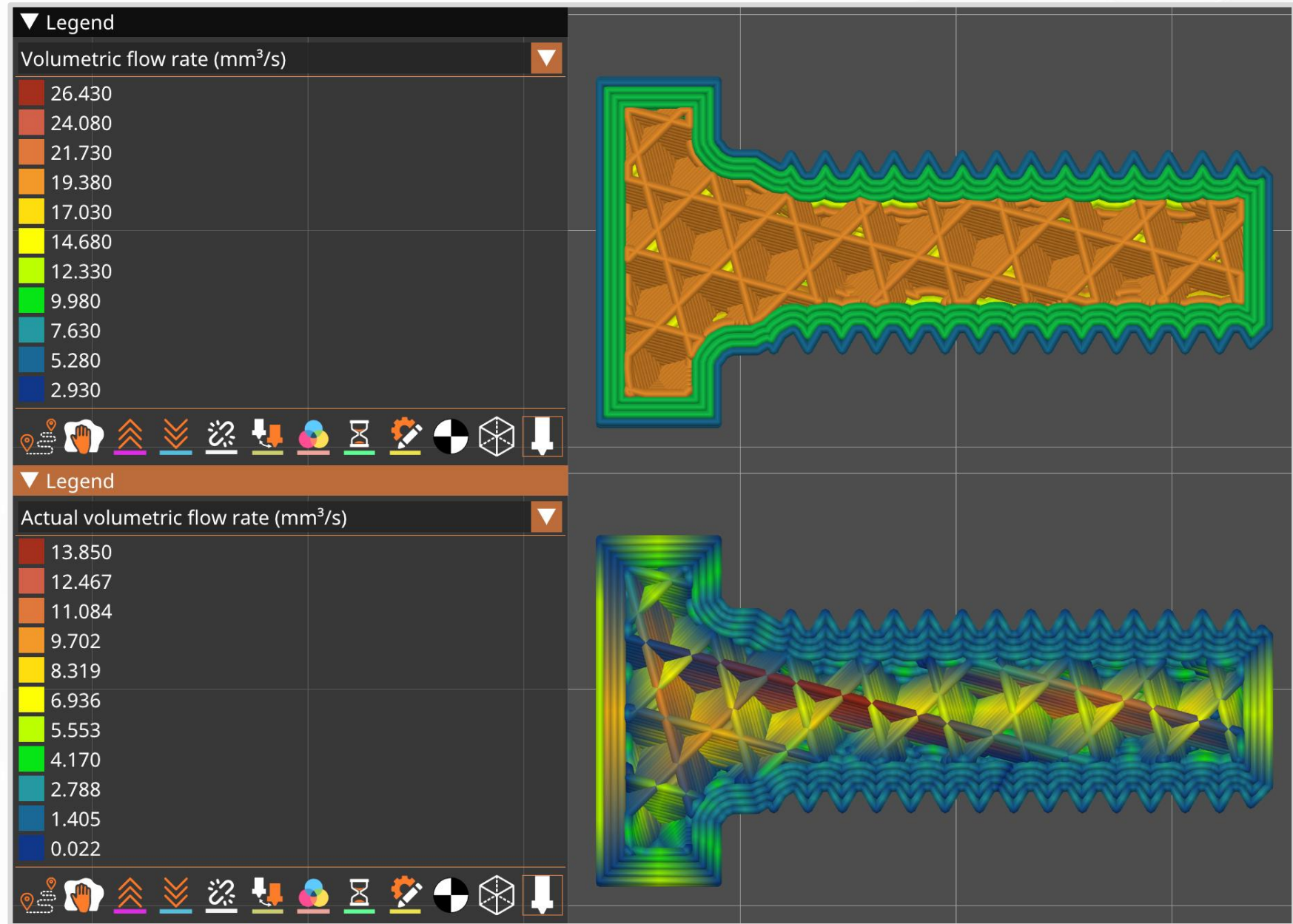
Why is the requested speed non-uniform?

- Adjust speeds according to geometry and visibility.
- Print slow on the outside to make pretty extrusions.
- Go fast on the inside to save time.



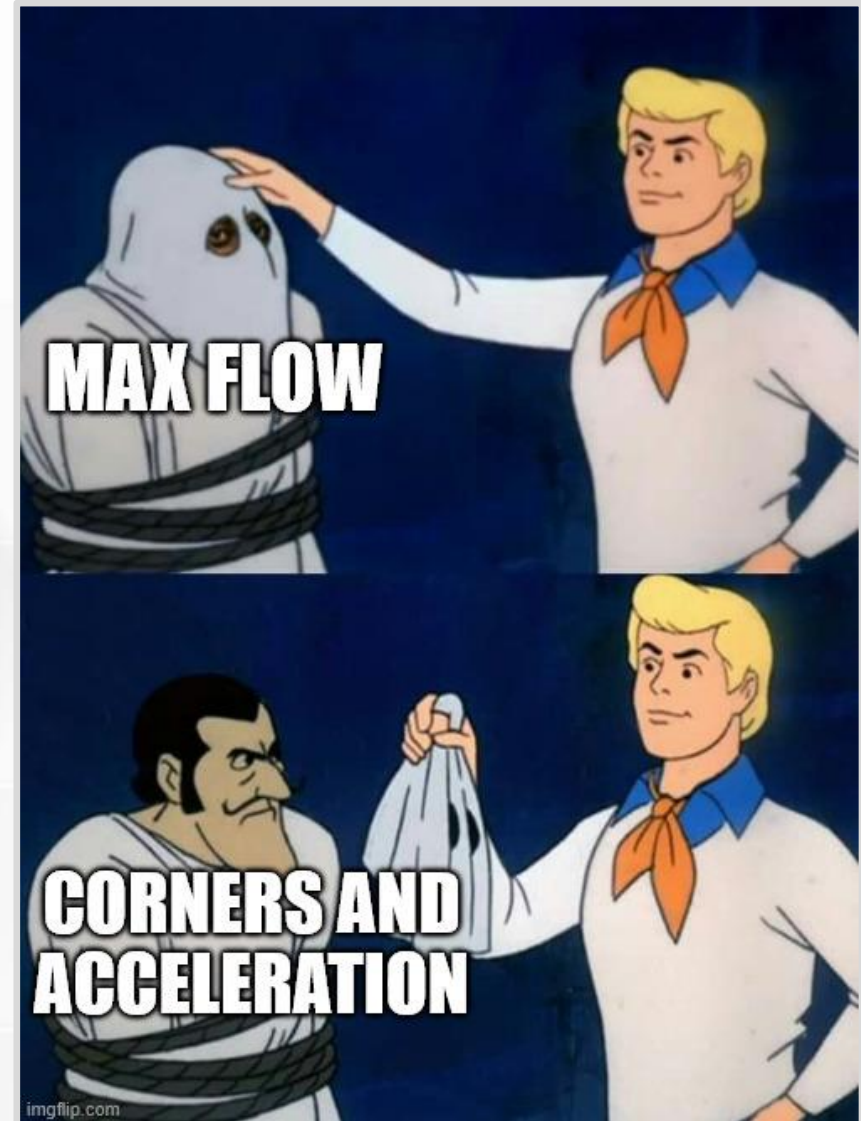
- Max requested flow:
22 mm³/s

- Max actual flow:
14 mm³/s



Speed Limits

- What is going on?
- My printer does not hit max speed, or max flow.
 - Max actual speed: $184 < 264$ mm/s
 - Max actual flow: $14 < 22$ mm³/s
- Limited by corners (geometry) and acceleration.



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Thermal Expansion

- Some materials expand as they heat.
- Coefficient of linear expansion.
- *Example:* Bridges use expansion joints to prevent gaps and buckling.
 - Between 1 inch and 1 foot.
- Is thermal expansion important for tiny 3D printer nozzles?

(Photo by Matt H Wade on [Wikipedia](#).)

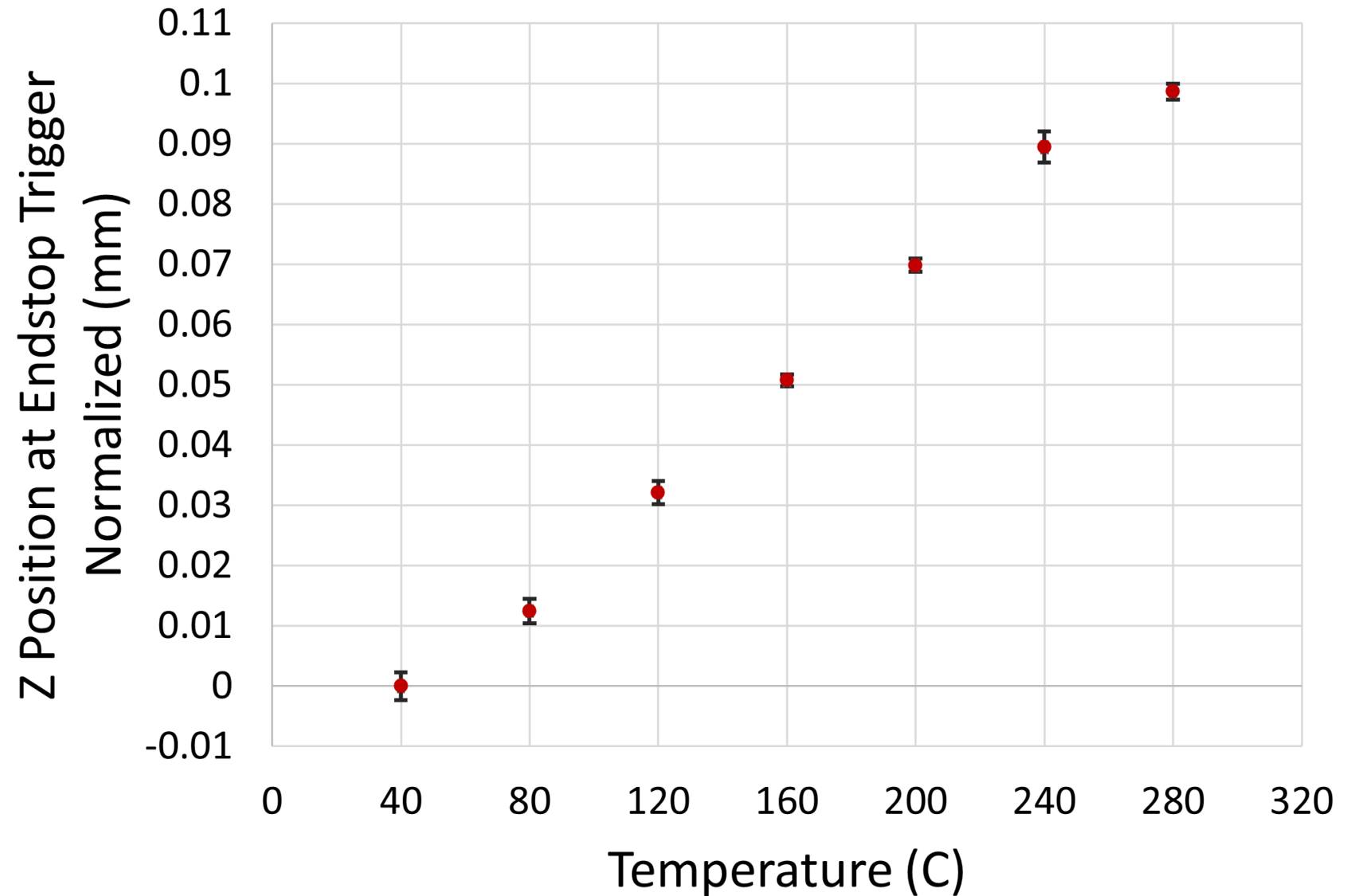


Thermal Expansion

- How much length does a hot brass nozzle gain?
- Simplified estimate with a 12 mm cylinder.
 - $\Delta T = 150\text{ }^{\circ}\text{C} \rightarrow + 0.034\text{ mm.}$
 - $\Delta T = 300\text{ }^{\circ}\text{C} \rightarrow + 0.068\text{ mm.}$
- Is this large enough to affect the first layer?
 - **Yes, first layers are very sensitive.**
 - Well known from experience.

- Data from my own printer.
- Set temperature and wait 10 minutes.
- 19 samples per point.
- My printer uses two probes and calculates an offset → Normalized to zero.
- Error bars $\pm$ one standard deviation.

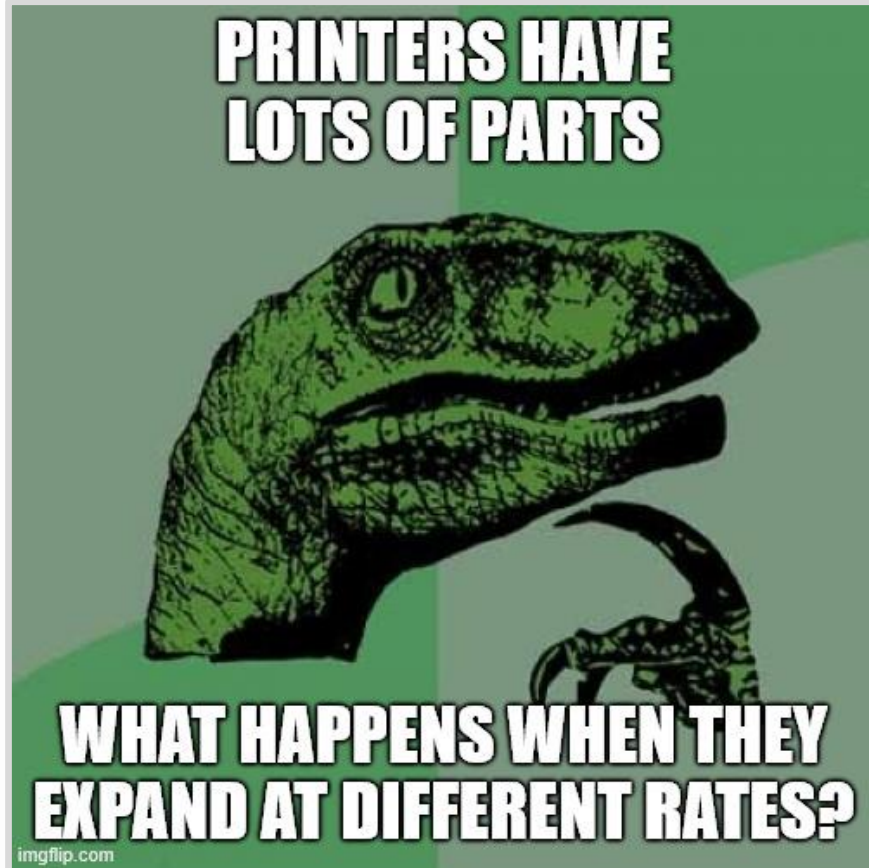
Thermal Expansion of a Brass Nozzle



Thermal Expansion

- What can you do about it?
 - “Heat soak.” Let everything heat up, expand, and wait for equilibrium.
 - Insert Gcode wait command.
 - Then probe nozzle position.
 - If you skip waiting, first layer is over-squished.
- Print beds need the same treatment.

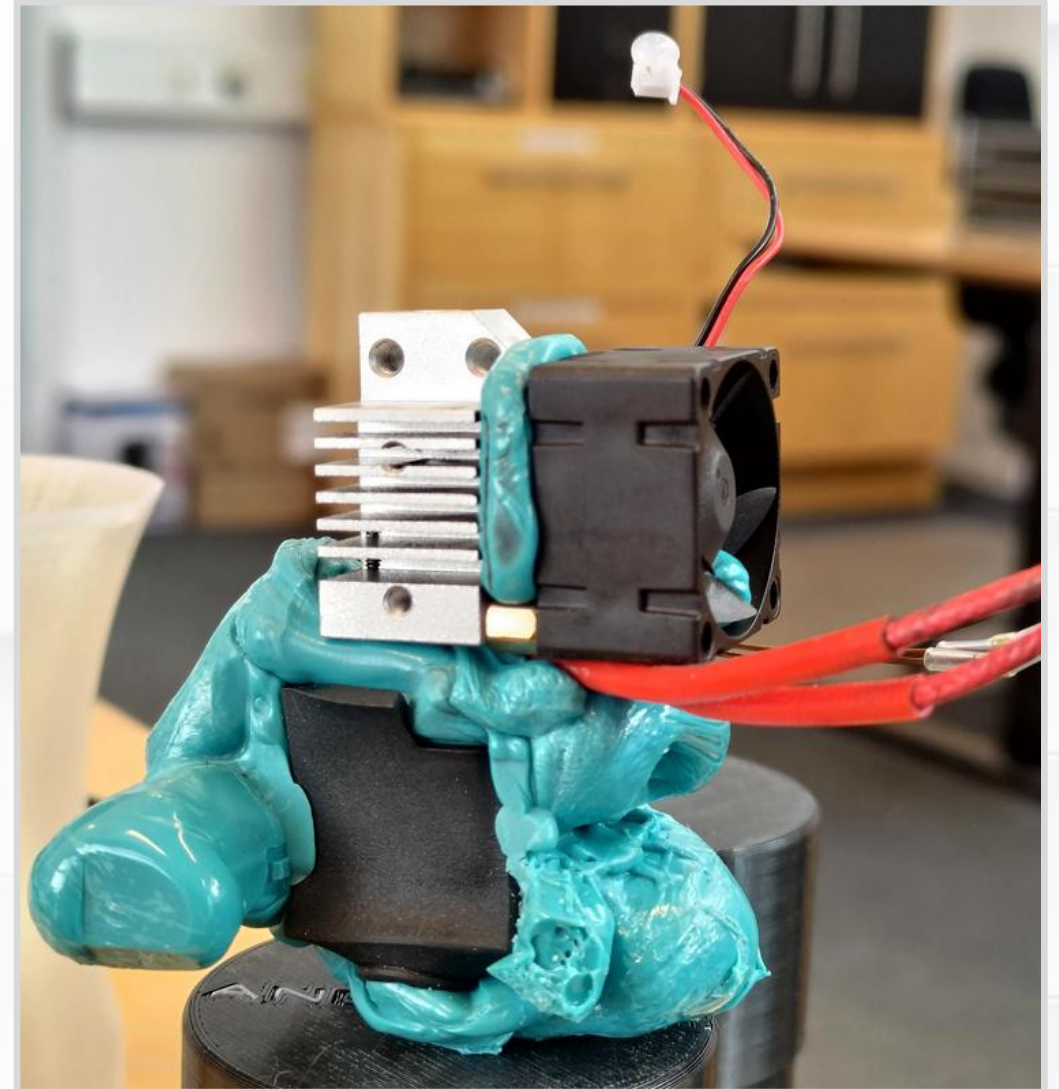
Thermal Expansion



Thermal Expansion

- Aluminum expands more than brass.
- Imagine you screw a cold brass nozzle into a cold aluminum heater block.
- What can go wrong?
- Unequal expansion at hot printing temps.
- Threaded hole expands faster than nozzle → A tiny gap forms in the threads.
- High pressure from extruding → Filament leaks out.
- “Blob of doom.”

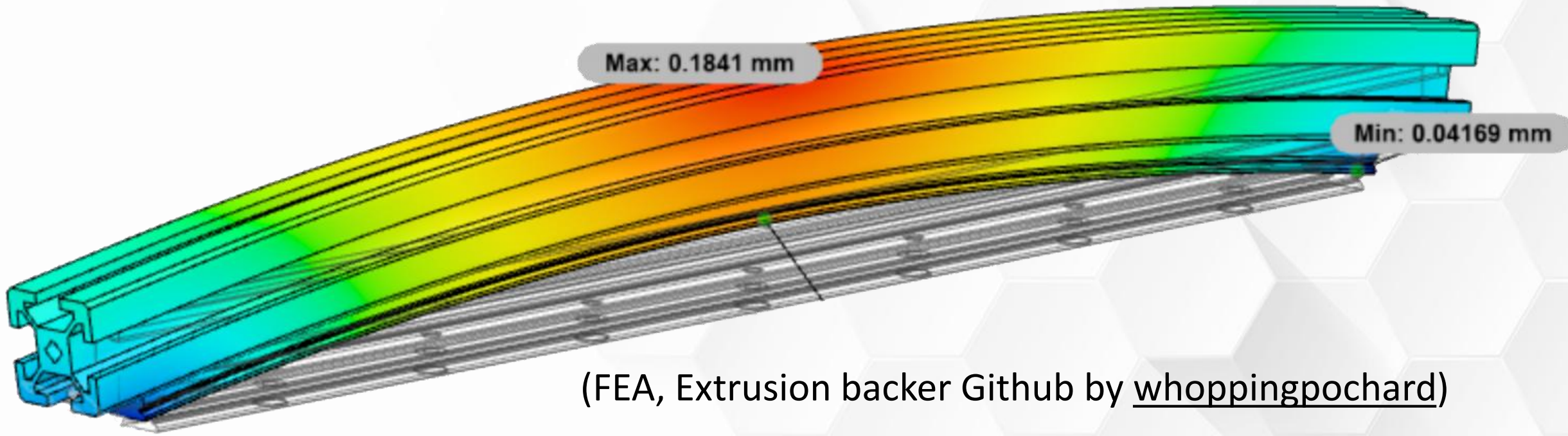
([reddit post.](#))



Bimetallic Strips

Consider the Y-axis from a Voron 3D printer.

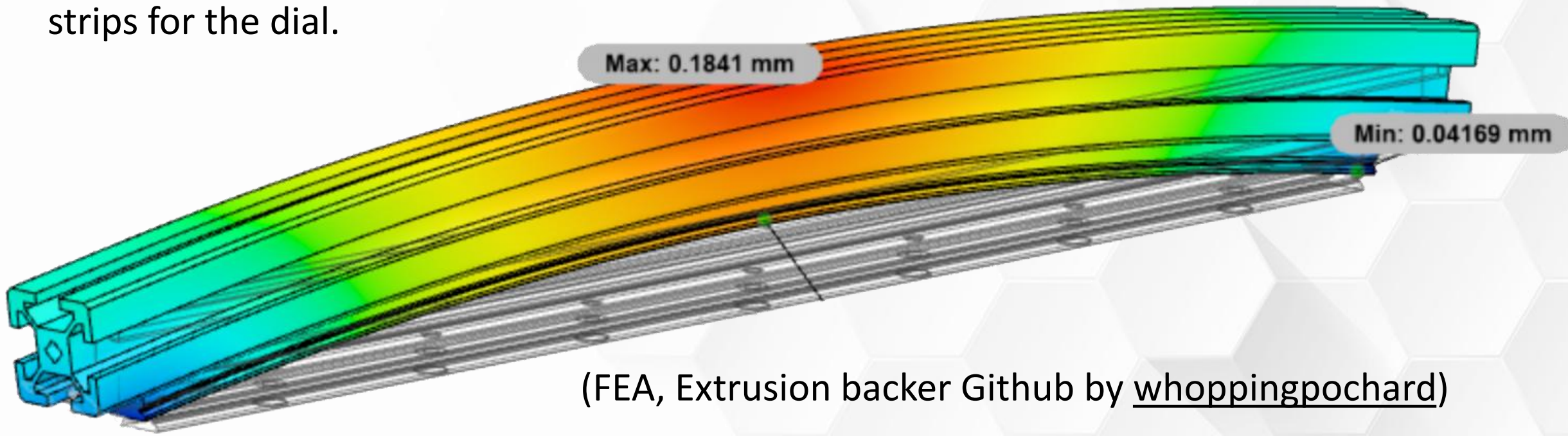
- Aluminum extrusion frame on top.
- Stainless-steel linear rail on bottom.
- Form a bimetallic strip (bar) when screwed together.



(FEA, Extrusion backer Github by [whoppingpochard](#))

Bimetallic Strips

- If assembled at room temperature, they start to bow as temps increase.
- Simulation for 300 mm beam, $\Delta T = 40^\circ\text{C}$. Almost 0.2 mm deflection.
- Fixed by adding reinforcing plate to the top.
- *Bonus fact:* Old fashioned thermometers use bimetallic strips for the dial.



(FEA, Extrusion backer Github by [whoppingpochard](#))

THANK YOU